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AI-Driven Approach for Automated Real-Time Pothole Detection, Localization, and Area Estimation

Younis Matouq[1], Dmitry Manasreh[1], and Munir D. Nazzal[1]

Abstract

Given the potential hazards and risks that potholes pose to road users, this study introduces an image-based system that utilizes a combination of a camera and GPS for real-time detection, georeferencing, and area estimation of potholes. The captured system data is processed in real-time using YOLOv8, a deep learning model proficient in object detection and segmentation. To enhance the precision and reduce the occurrence of false detections, the system is specifically trained to detect potholes, manholes, and patches. Additionally, the camera is calibrated to accurately estimate the area of identified potholes. The proposed system achieved a mean average precision of 91% in detecting potholes, 98% in detecting manholes, and 90% for detecting patches. A salient feature of this system is its capability to localize potholes with reference to pavement line lane markings. This ability could facilitate proactive lane closure planning by maintenance crews, further enhancing road safety measures. The study findings suggest that the system holds significant potential for practical implementation. Its deployment could assist transportation agencies in the prioritization of road repairs, resource allocation, and advance planning for lane closures, ultimately enhancing the efficiency of their maintenance workflows.

Keywords

infrastructure, infrastructure management and system preservation, maintenance management systems, asset inventory, data collection

Potholes have been recognized and defined by numerous researchers in the field of pavement engineering, with each definition shedding light on different causes of their formation. Chatterjee et al. highlighted the mechanical process of pothole formation, emphasizing how water infiltration and traffic loads lead to the disintegration of the asphalt surface layer (*1*). This is in alignment with the definition provided by Griffith and Maher et al., who identified potholes as localized areas of rapid asphalt pavement deterioration, resulting in the loss of paving material and the creation of steep depressions (*2*, *3*). Wang et al. expanded on this by considering the climatic conditions that contribute to pothole formation, such as freeze-thaw cycles and rainfall, and their interaction with existing pavement defects (*4*). Nazzal et al., Yuan et al., and Maher et al. have underscored the impact of potholes on traffic safety and infrastructure management, highlighting the economic and safety consequences of their formation (*3*, *5*, *6*). As such, they represent a key concern in the field of pavement engineering and road maintenance.

Manually evaluating the condition of pavements requires a significant amount of training and skill. In addition, it is impractical for an individual to manually evaluate every mile of a nation's road network within a given time frame. Not only is this method demanding, but it is also likely to suffer from inaccuracy. To optimize this procedure and improve its efficacy, cost-effectiveness, and dependability, researchers have been examining automated methods for evaluating the condition of road surfaces. These methodologies frequently employ techniques from computer vision, machine learning, and deep learning.

[1]Department of Civil and Architectural Engineering and Construction Management, Center for Smart, Sustainable and Resilient Infrastructure (CSSRI), University of Cincinnati, Cincinnati, OH

Corresponding Author:
Munir D. Nazzal, nazzalmd@ucmail.uc.edu

Literature Review

On the basis of conventional image segmentation, Koch and Brilakis sought to optimize and refine these techniques (*7*). They developed a technique for the automatic detection of potholes in asphalt pavement images. In their suggested approach, the image was initially segmented into defective and non-defective regions based on the threshold processing of the histogram's shape. The geometric properties of the defective region were investigated further, with morphological refinement and elliptical regression used to approximate pothole shapes. This method's ability to precisely recognize potholes in asphalt pavement images demonstrates its effectiveness.

In their 2020 study, Gao et al. proposed an innovative method for accurate pothole segmentation that combined the processing of grayscale and texture features with a machine learning model based on the library of support vector machines (LIBSVM) (*8*). The research made use of various image processing techniques and an industrial camera to identify potholes rapidly and precisely. Existing image processing techniques for pavement distress detection were outperformed by the accuracy, precision, and recall of the experiment results. The researchers emphasized that the superiority of their method resulted from the combination of grayscale and texture features for pothole segmentation, as opposed to using either characteristic alone.

The major benefit of deep learning is its minimal need for user involvement, making it an attractive choice for researchers across various disciplines. Its wide array of applications includes object recognition, categorization, and partitioning (*9*).

Ping et al. introduced a system for detecting potholes, which was trained on a dataset comprising 2,036 images captured using a smartphone placed on a car's dashboard (*10*). Their research involved comparing the effectiveness of four distinct models: You Only Look Once version 3 (YOLOv3), Single Shot Detector (SSD), Histogram of Oriented Gradient (HOG) with Support Vector Machine (SVM), and Faster Region-based Convolutional Neural Network (R-CNN). They discovered that YOLOv3 demonstrated superior performance, achieving an accuracy of 82%. Although the researchers asserted that the dimensions of the identified potholes could be measured, they did not provide supporting results for this claim. In concluding their study, they emphasized the significance of detecting manholes covers along with potholes in future research.

In their study, Kharel and Ahmed introduced a system designed to identify potholes and quantify the area of the detected potholes (*11*). To do this, they employed Mask R-CNN for pothole detection and instance segmentation. The training of the model was accomplished using 240 images, while 51 images were used for validation. The researchers determined the pothole's area by isolating the pixels that represented the pothole and multiplying them by the square centimeter size of a pixel. The pixel size was measured based on area of the road that each image covered and the resolution of the image. However, a notable limitation of this area-measurement approach was its assumption of a constant square centimeter pixel area throughout the image frame. This does not account for any perspective distortion.

Kharel and Ahmed designed a model for detecting potholes in real-time and estimating their area (*11*). They curated their dataset from Google Images, using 665 images for model training and 183 images for validation. Their study compared the performance of six convolutional neural networks (CNNs): three variations of YOLOv5 (Yl, Ym, and Ys) and Faster-RCNN with three different backbones (ResNet50, VGG16, and MobileNetV2). Because of its impressive speed, they suggested using Ys for real-time applications. The area of the identified potholes was estimated by obtaining a top-down view of the bounding box using inverse perspective mapping, calculating the polygon area in pixel2, which resulted from this process, and then converting it to actual measurements using a fixed scale. However, this method has certain limitations, including the assumption of a constant pixel size within the image frame, and the calculated area being representative of the bounding box area, which is larger than the actual pothole area.

Heo et al. introduced a technique for detecting potholes and assessing their risk (*12*). They utilized a publicly accessible dataset comprised of 665 images, applying data augmentation methods to expand the image count to 2,665. Then, 70% of this augmented data was used for training. The researchers assessed the effectiveness of four distinct models: YOLOv2, YOLOv3, YOLOv4 tiny, and SPFPN-YOLOv4 tiny. The SPFPN-YOLOv4 tiny model demonstrated superior performance, achieving a mean average precision (mAP) of 79.6%. Assuming a monocular camera, the researchers estimated the distance between the camera and the pothole. They then applied the pinhole camera formula to determine the real size of the pothole from the detected bounding box. Ultimately, they categorized the potholes as either safe or dangerous by comparing their size to the tire contact patch size.

Bučko et al. conducted a study to determine how adverse weather conditions affect the efficiency of pothole detection. In varying weather conditions, the researchers compared the performance of two models, YOLOv3 and Sparse R-CNN (*13*). The dataset used to train the models was collected from the same section over the course of 3 months, from May to July. The dataset included 2,099 images captured at various times of day and under varying weather conditions. Because of their high similarity, the authors chose to classify manholes alongside potholes in an effort to reduce false positives.

They discovered that YOLOv3-SPP outperformed Sparse R-CNN, achieving a mAP of 0.771 at a 50% confidence level.

The existing literature pertaining to image-based pothole detection systems highlights several areas that remain to be addressed. The initial apparent concern revolves around the precision of estimating the area of potholes. While numerous systems possess the capability to identify potholes, they frequently encounter difficulties in precisely determining their area.

Another notable deficiency lies in the composition of the data used for training the detection models. The limited amount of training data in the current literature can result in models that exhibit poor generalization when applied in real-world situations, thereby affecting their practical efficacy. Adding to the complexity of this matter is the observation that the majority of image-based algorithms that concentrate on the detection of potholes fail to sufficiently consider the notable resemblance shared by potholes, manholes, and patches. Therefore, there exists an urgent need for enhanced training datasets and methodologies that effectively tackle these challenges. Finally, many studies fail to provide comprehensive case studies that illustrate the practical effectiveness of their proposed models, consequently restricting their adoption and full potential utilization.

Objective and Scope

The goal of this study is to propose and develop a practical and economical real-time automated system for detecting potholes. This system utilizes advanced machine learning algorithms, especially YOLOv8, along with a robotic operating system (ROS) to automate the process. The system is designed to detect, georeference, and estimate the area of potholes across a wide range of real-life conditions. In addition, the system provides a precise localization of detected potholes with respect to pavement lanes by identifying the location of pavement lane markers.

Methodology

Development of a Real-Time Detection System

To address the complexities involved in manually detecting and georeferencing potholes, a system leveraging a set of artificial intelligence (AI) algorithms was developed, with YOLOv8 being the core algorithm. This system is comprised of two main components—hardware and software—both of which will be elaborated on in detail in this section.

A specific part of this section has been devoted to detailing the methodologies used to collect data, identifying the classes used for training, and providing an in-depth description of the final dataset utilized for training the core algorithm.

To enable a precise location reporting of the detected potholes, an instance segmentation model was employed to detect and segment line lane markings. The pothole location is subsequently reported with reference to these detected lane markings. This section will provide a full description of the methods employed to annotate, train, and incorporate the detection outputs of both models, with the goal of achieving precise localization of the detected potholes. In an effort to estimate the area of the detected pothole, a calibration experiment was carried out on the camera to gain a rough estimate of the real area corresponding to each pixel in the image frame. The data obtained from this experiment was subsequently modeled using two different approaches. A detailed description of the experiment, the techniques used for data modeling, and the underlying assumptions and basis on which the experiment was conducted will be explored within this section.

System Description

Two sensors were utilized in the development of this system: a camera and a GPS. The camera was initially mounted on an angle-adjustable tripod with a 32-degree downward tilt from the horizontal plane, followed by mounting both sensors to aluminum extrusions. These extrusions were then horizontally positioned on a ladder rack, which was itself mounted on a truck bed. This configuration provided flexibility and portability. The camera had a sensor size of 10.36 mm, a resolution of 2,880 $\times$ 1,860 pixels, an exposure time ranging from 86.128 μs to 5 s, and a frame rate of 20.8 frames per second (FPS). The resolution was cropped to a resolution of 1,920 $\times$ 780 pixels to serve the purpose of this study. To prevent capturing blurry images, mainly caused by the fast movement of the vehicle, the exposure time was set to 1,200 μs. The frame rate used was 12 FPS. Because of its wide field of view, a 3.5 mm C Series fisheye lens was used, enabling the capture of two lanes with high quality. Lastly, specialized cables were used to connect the sensors to a laptop.

The system facilitated ROS, which is a flexible framework designed for the development of software for robotic applications. It employs a distributed, modular system that allows various software processes, known as nodes, to communicate with one another. The main algorithm used for detecting and segmenting potholes was YOLOv8, which is contained within a node termed as the pothole detection node. This node subscribes to the rectified image and the latitude and longitude coordinates from the GPS. Once an image is received, it is processed by the pothole detection node. If a pothole is

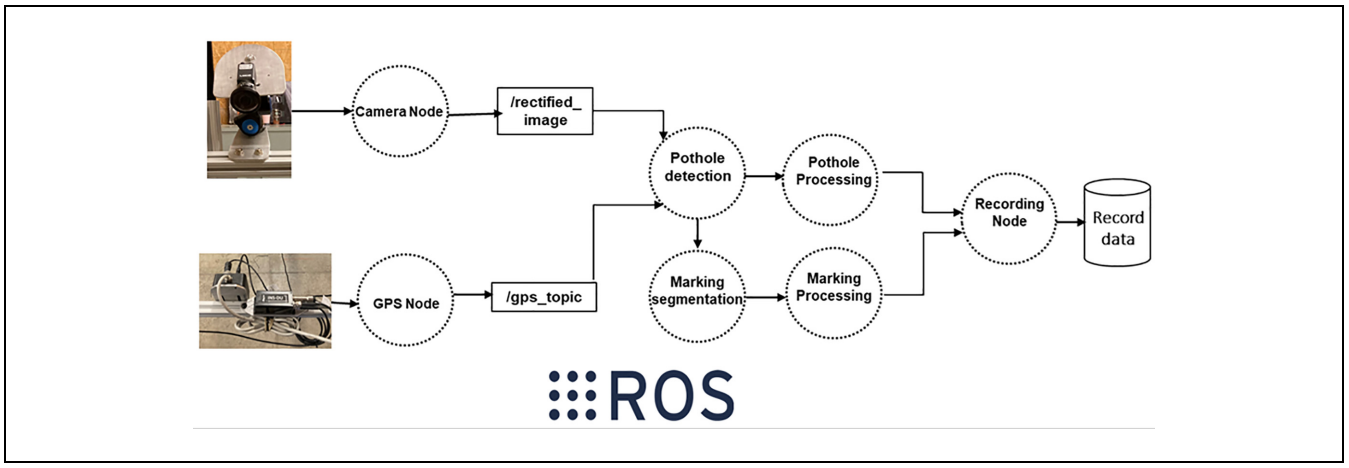


Figure 1. System architecture diagram.
Note: GPS = global positioning system; ROS = robotic operating system.

detected, the pothole detection node publishes the rectified image, the binary image masking the detected pothole, the center coordinates of the detected pothole in the image, and the latitude and longitude coordinates. This data is subsequently processed using multiple nodes to obtain the estimated area of the detected pothole, and the location of the pothole with respect to pavement lanes. The flowchart in Figure 1 shows the system architecture; further details of the system processes will be discussed in the following sections.

Data Collection and Dataset Preparation

Data was collected from various locations within the state of Ohio, U.S., over a span of 8 months to train a detection model. The aforementioned system was used for data acquisition. Despite the primary objective of this study being to detect potholes, the presence of certain objects on the pavements with a high degree of similarity to potholes could lead to poor model performance if not accounted for. Therefore, the data was annotated into three distinct classes: potholes, manholes, and patches. Patches and manholes were chosen because of their high similarity to potholes. The dataset consisted of 23,661 unique images, which are further divided into background images, totaling 14,247, and images with objects, totaling 9,414. The distribution of labels across all the images is depicted in Figure 2. The total number of images containing objects and backgrounds is shown in Figure 3.

To avoid biases during data collection and to ensure that the collected data is representative of the population being modeled, data was collected during different times of the day and from a variety of locations across Ohio state. The times at which pothole data were collected and the locations covered in this study are illustrated in Figures 4 and 5, respectively.

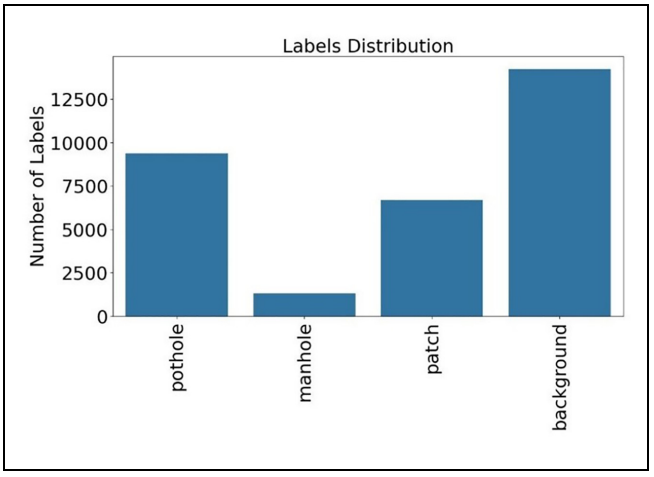


Figure 2. Label distribution for the potholes dataset.

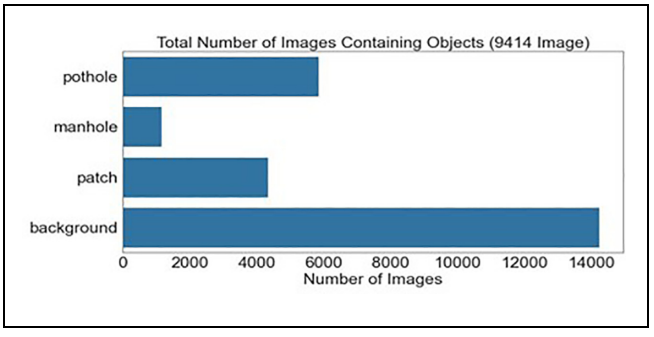


Figure 3. Total number of images containing objects and background for the potholes dataset.

One of the main objectives of this system is to perform real-time detection and instance segmentation on the potholes; however, annotating the data for a segmentation task is time-consuming and labor-intensive. For the purpose of converting rectangular annotations to segmentation annotations, an algorithm was developed, which will

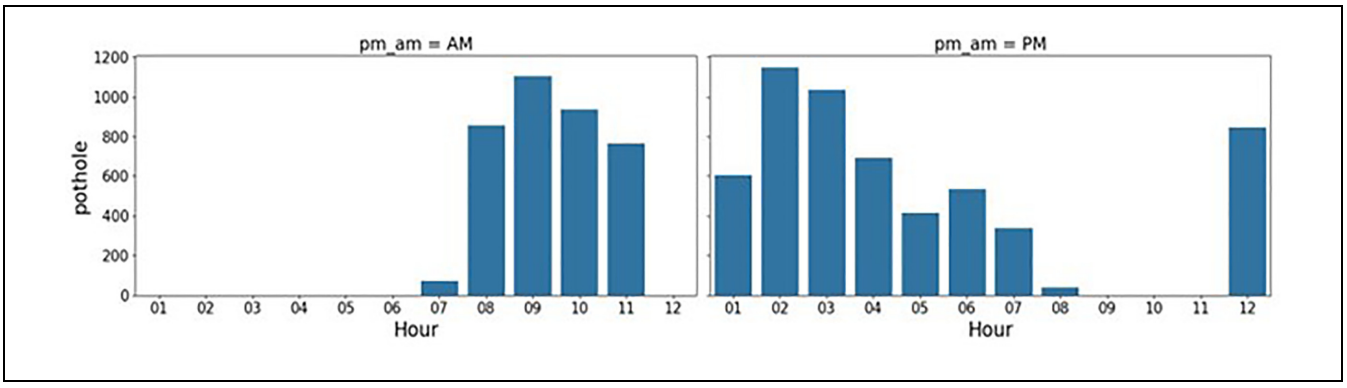


Figure 4. Time of potholes dataset collection: a.m. (*left*) and p.m. (*right*).

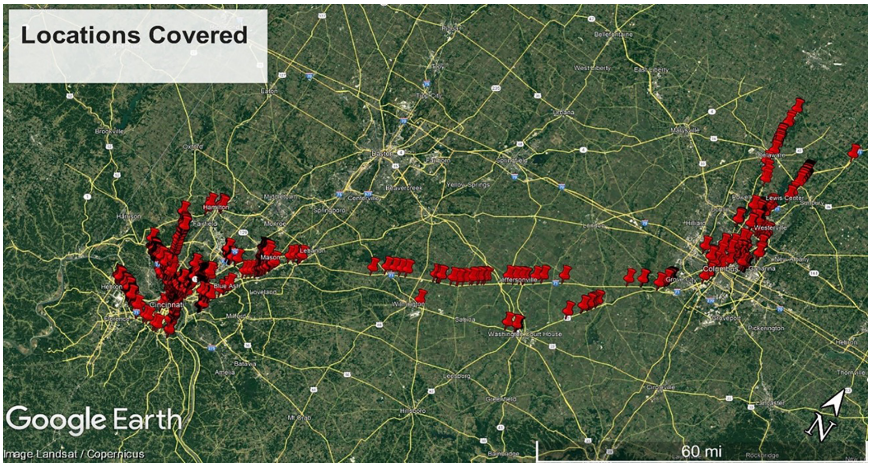


Figure 5. Locations covered during potholes data collection. The red pins are locations potholes covered during this study.

be referred to as the "annotation-converter" algorithm. The annotation-converter primarily uses the Segment Anything Model (SAM), a deep network published by the Meta AI research team (*14*). This model can segment any image given to it, or any specified portion of an image, and returns a binary image containing masks of the segmented objects; however, discussing SAM in detail is beyond the scope of this study.

An overview of how the annotation-converter algorithm operates is as follows: the algorithm receives an image and its corresponding rectangular annotations as input, the input is passed to SAM and the binary image with the masks is generated, then the algorithm uses a function to find the contours within the masks generated by SAM, and, using these contours, the polygon's vertices will be approximated; these vertices represent the segmentation annotations. The instance segmentation YOLOv8 model was trained using these annotations. The input and output of the annotation-converter algorithm are illustrated in Figure 6.

Training of Instance Segmentation Model for Potholes, Patches, and Manholes

The YOLO model, among the range of object detection algorithms, has distinguished itself for its impressive blend of speed and accuracy. As a result, it offers fast and accurate detection of objects in images. Since its initial development, improvements have been made to successive iterations of the YOLO family, with shortcomings being addressed and capabilities being enhanced (*15*). In January 2023, the YOLOv8 version was unveiled by Ultralytics, with its applications being expanded to include object detection, segmentation, pose estimation, tracking, and classification (*16*). The YOLOv8 model has been selected for the task of detecting and segmenting potholes within this system.

The modified CSPDarknet53 backbone is incorporated into the architecture of YOLOv8, with the C2f module being the main component of its structure. The accuracy of detection is enhanced by integrating high-level features with contextual data in this module. The

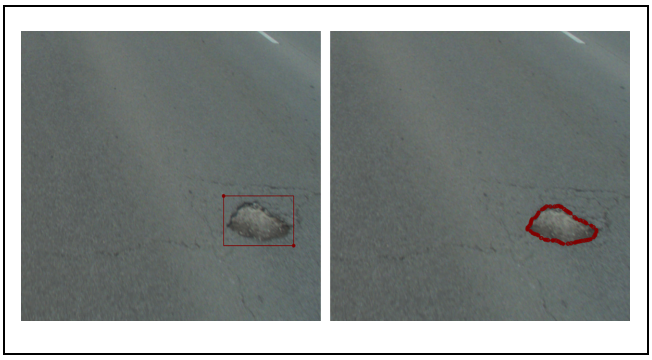


Figure 6. Annotations converter algorithm input (*left*) and output (*right*).

utilization of an anchor-free model, coupled with a decoupled head, allows YOLOv8 to independently handle objectness, classification, and regression tasks. This enables each branch to concentrate on its respective task, resulting in a boost in the model's accuracy. In the output layer of YOLOv8, the sigmoid activation function is used for the objectness score, which denotes the probability of a bounding box enclosing an object. The softmax function is used to calculate class probabilities, which represent the likelihood of objects belonging to each possible class. With regard to loss functions, YOLOv8 employs complete intersection over union and distributed focal loss for bounding box loss, and binary cross-entropy for classification loss. The beneficial enhancement of object detection performance, especially with smaller objects, has been proven by these loss functions (*15*).

The model was trained to perform instance segmentation, and the training process was initialized from scratch without the incorporation of any pretrained weights. The division of the previously detailed dataset into training and validation sets was undertaken, with the former constituting 75% and the latter 25%. Training of the model persisted until convergence was reached, and the best weights that yielded best results on the validation set were logged. Stochastic gradient descent was adopted to optimize the model's weight with the initial learning rate set to 1×10^{-2}. To avoid gradient explosion, a warm-up learning rate was applied during the first five epochs. Subsequently, a cosine learning rate scheduler was used to schedule the learning rate. The training process was facilitated using two NVIDIA A100 Graphics processing units (GPUs).

Line Lane Marking Detection and Pothole Localization

A model dedicated to the segmentation of line lane markings was developed, with an emphasis on three distinct classes: continuous white lines, dashed white lines, and continuous yellow lines. The primary motivation behind this model was to achieve the localization of detected potholes in relation to line lane markings, permitting the precise reporting of the georeference of the pothole and the specific lane in which the pothole is present. Given the time-consuming and labor-intensive nature of annotating images for instance segmentation, as previously stated, an unsupervised algorithm, here referred to as the lane-marking annotator, was conceived to expedite the annotation process.

To elucidate, the lane-marking annotator algorithm takes an image as input and yields the vertices of the polygons that represent the line lane markings, adhering to the YOLO annotation format for instance segmentation. The algorithm leverages image processing, a clustering algorithm, and a deep learning algorithm to accomplish this task. SAM is employed by the lane-marking annotator to isolate the objects of interest. As highlighted earlier, SAM can segment any part of an image, as it is a promptable algorithm. This implies that it can accept an image and coordinates and will then segment the image into foreground and background based on the given coordinates.

A critical component of the process is the unsupervised input of the coordinates of the objects of interest to SAM. To achieve this, the image is initially processed using a Gaussian blur filter, followed by a conversion from the Red, Green and Blue (RGB) format to Hue, Saturation, and Value (HSV). This step is designed to filter out the indices of the desired colors, specifically white and yellow, based on their ranges in HSV. Once the indices for these colors are identified, they are utilized as input for a k-means clustering algorithm, which has been specified to have six cluster centers. These cluster centers then serve as the prompt for SAM, which consequently segments the image into foreground and background. A binary image containing masks of the segmented objects is the output of this process. Functions to find contours and approximate the polygon vertices from these contours are then employed. These vertices are subsequently used as annotations. The flowchart in Figure 7 illustrates the model's input, output, and intermediate steps.

Using the lane-marking annotator algorithm, a dataset was created. This dataset, which consisted of 1,186 images, included three classes: white lane marking, yellow lane marking, and white dashed lane marking. Figures 8 and 9 show the total number of images used and how the labels were distributed among them, respectively. The dataset was split into training sets and validation sets, with 70% of the data for the training set and 30% for the validation set. A YOLOv8 model was then trained to perform instance segmentation.

Given that pothole localization was the main goal of training this model, the following workflow was established to achieve accurate localization of the detected potholes:

- The line lane markings are detected and segmented using the model.

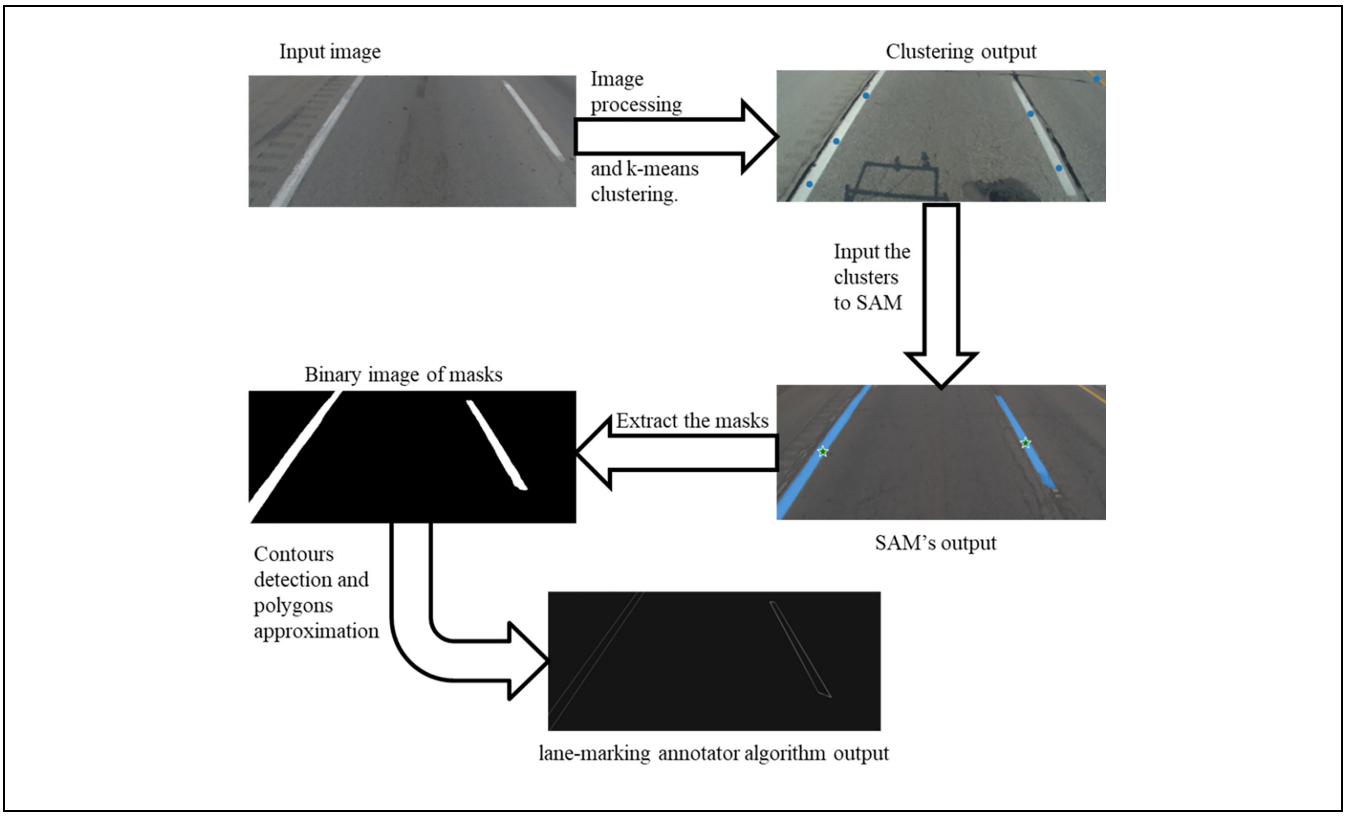


Figure 7. Lane marking annotator algorithm flowchart.
Note: SAM = Segment Anything Model.

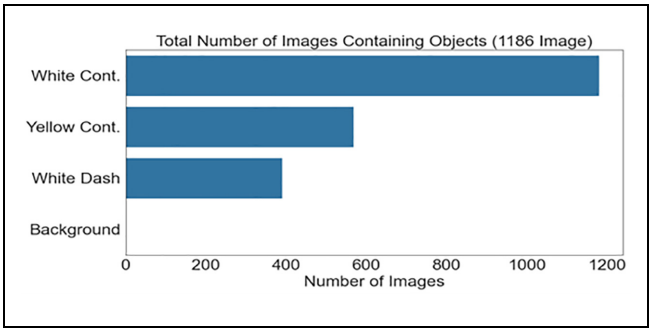


Figure 8. Total number of images containing objects and background for the line lane marking dataset.
Note: Cont. = continuous.

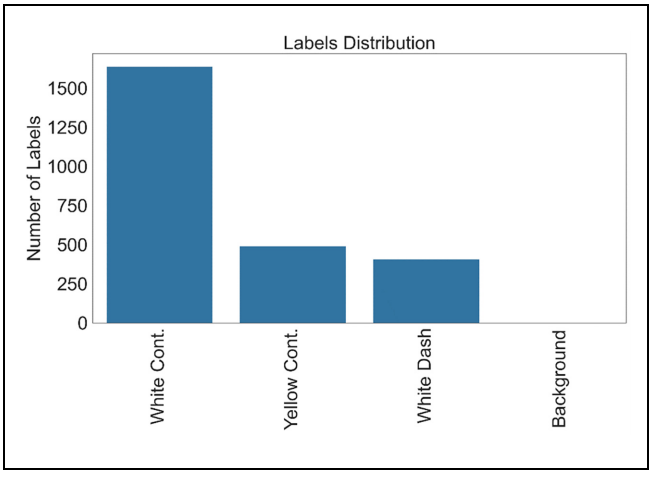


Figure 9. Label distribution for the line lane marking data.
Note: Cont. = continuous.

- These segmented line lanes are masked out from the image.
- A line fitting is applied to identify a line that traverses through the line lane center.
- The central line extended to reach both ends of the image using linear regression.
- The detected pothole is then localized with reference to the line lanes.

The flowchart in Figure 10 illustrates the process of detecting and processing the line lane markings.

Camera Calibration for Potholes Real Area Assessment

A methodical experiment was carried out to obtain a rough estimate of the actual area of a detected pothole using the segmented mask. However, it should be noted that the results obtained from this experiment are strictly applicable under the conditions in which it was executed. The applicability of the results depends on several conditions. Firstly, the camera should be angled 32 degrees

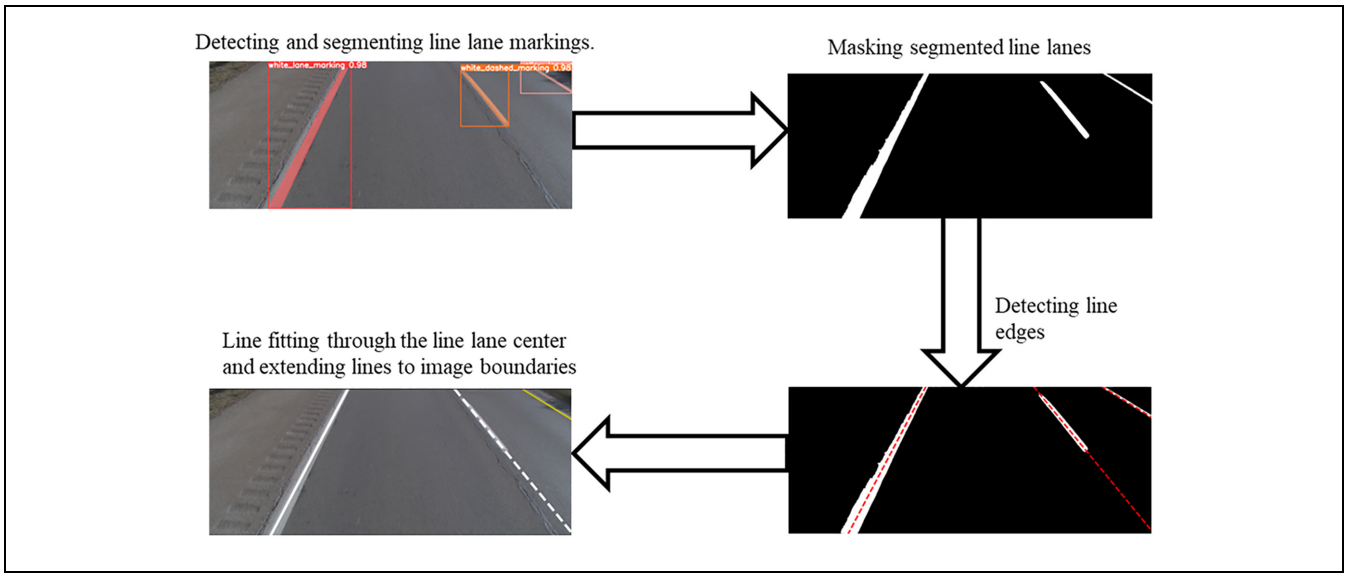


Figure 10. Process of detecting and processing the line lane markings.

downwards from the horizon. Secondly, the elevation of the camera from the ground must be 87.5 in. Lastly, the camera parameters should correspond to those utilized during the experiment.

To simulate the system's anticipated operational environment, a paved parking lot was selected as the experimental site. The system was initially mounted on a truck bed, and then the boundaries of the camera's field of view were marked out on the pavement. A grid was created within these boundaries. Then, at each grid point, a two-dimensional object of known height and width dimensions was placed, and an image was captured. The experiment was conducted in two phases. In the first phase, three boards of identical size but varying colors were used as reference objects, each measuring 22×28 in. During the second phase, two boards of different sizes and colors were employed as reference objects, measuring 8.3×11.7 in. and 16.6×11.7 in.

In the first approach, to assess the corresponding area, the real area of the board was divided by the number of pixels representing it. This value was then assigned to the pixels corresponding to the board. In instances of overlap, the values assigned to the overlapping pixels were replaced with the average of the overlapped pixel values. Figure 11 shows the overlapping between two boards.

At the end of this process, an image was generated where each x and y coordinate was associated with the real-world area. This area was then utilized to map the mask of the detected pothole to the real area.

In the second approach, the problem was addressed as a regression task, with the image serving as the input and the real area of the boards within the image considered as the target. To train this data, a deep CNN, ResNet, followed by a fully connected layer that outputs a

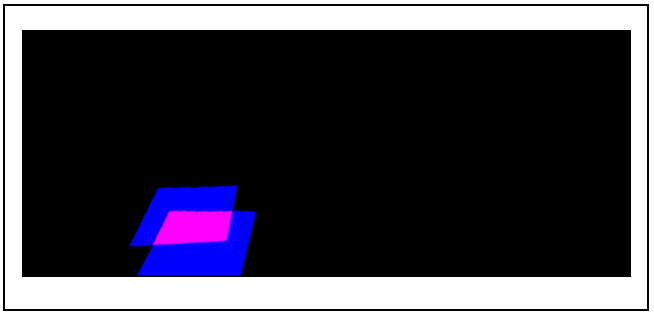


Figure 11. Overlapping between two boards.

continuous number, was employed (*17*). A backpropagation algorithm was used to train the model. Mean squared error loss was used to measure the difference between the actual area of the board and the predicted value of the model. Subsequently, stochastic gradient decent with a learning rate of 1×10^{-5} was used to optimize the model's parameters. Figure 12 depicts the experiment and system setup.

Results and Discussion

Potholes Instance Segmentation Model

The YOLOv8 instance segmentation model developed for the detection of potholes, patches, and manholes, converged after 150 epochs. The model's performance was primarily evaluated based on the mAP at a 50% confidence level. mAP is a commonly used metric for evaluating the performance of object detection models. Calculating the average precision (AP) across all categories provides a single value that facilitates the comparison of various models. The mAP metric is founded on

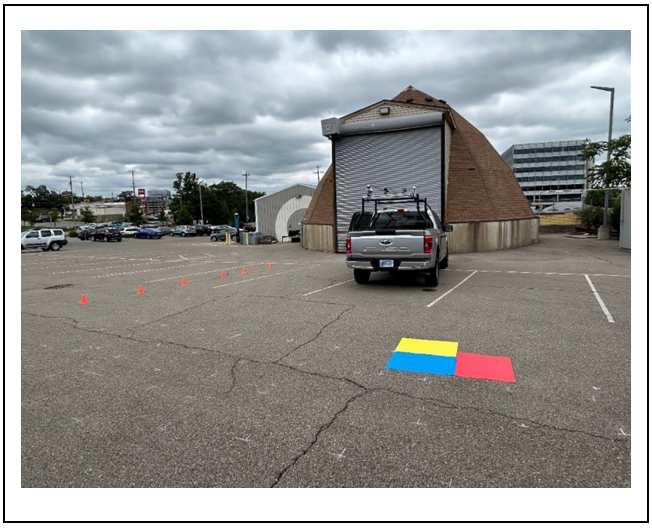


Figure 12. The experiment and system setup.

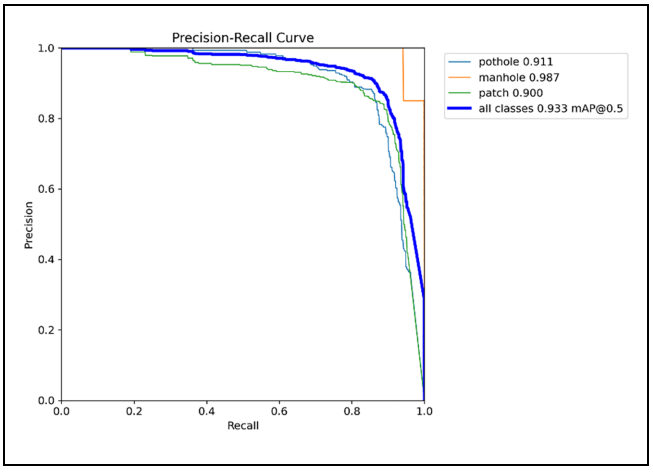


Figure 13. Precision-recall curve for pothole detection model.

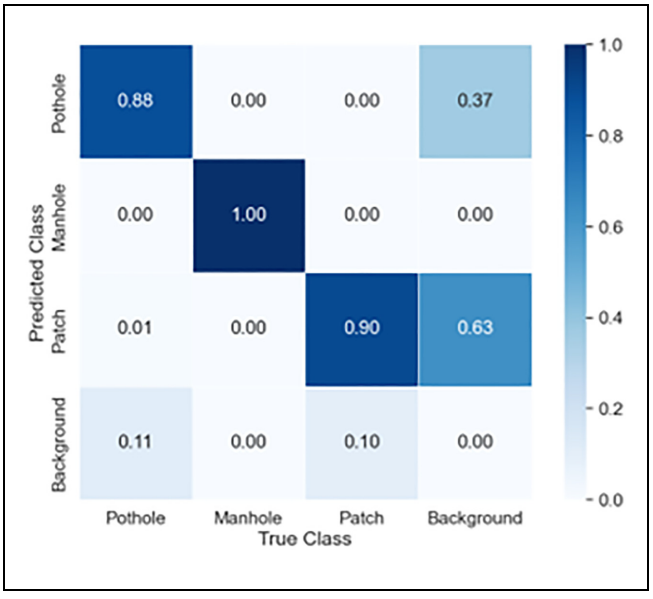


Figure 14. Pothole model confusion matrix.

precision-recall metrics and defines positive prediction using intersection over union (IoU). The IoU is calculated by dividing the intersection region by the union region of the predicted and actual bounding boxes. Precision and recall are fundamental aspects of this metric. Precision measures the accuracy of a model's positive predictions, whereas recall quantifies the proportion of true positive cases that the model correctly identified. Given the typical trade-off between precision and recall, the mAP metric employs the precision-recall curve, which plots precision against recall for various confidence thresholds. By considering the area under the precision-recall curve, this method provides a balanced assessment of both precision and recall. Additionally, the mAP metric addresses the need for models to accurately identify and locate multiple object categories within an image. To ensure a comprehensive evaluation of the model's performance, the model calculates the AP for each category separately and then the mean of these APs across all categories. As illustrated in Figure 13, the precision-recall curve showed that the model achieved a mAP of 91% for pothole detection, and 98% and 90% for patches and manholes, respectively.

In a test with 318 images containing 695 instances (including 290 potholes, 37 manholes, and 368 patches), the confusion matrix, shown in Figure 14, highlighted the model's robust performance. It successfully identified 88% of actual potholes, with 11% undetected (classified as background), and a mere 1% misidentified as patches. All manholes, owing to their consistent shape and texture, were accurately classified, and the model correctly detected 90% of patches, with the rest misclassified as background. Background detection, or false positives, were minimal, totaling 32. Of these, 63% were misidentified as patches, and 37% as potholes. This demonstrates the model's resistance to false positives, which was enabled by the significant amount of background images used during training. Furthermore, the model demonstrated a strong ability to differentiate between potholes, patches, and manholes, ensuring its high validity in real-world scenarios. The robustness against false negatives can be attributed to the comprehensive and clean dataset used for training. This dataset, collected systematically over 8 months and at various times of the day, simulated real-world conditions, thereby enhanced the model's overall performance. Figure 15 depicts the model's detection and instance segmentation.

Line Lane Marking Instance Segmentation Model

The YOLOv8 instance segmentation model developed for the detecting line lane marking, converged after 60 epochs with a 70% mAP @ 0.5 for all classes. The

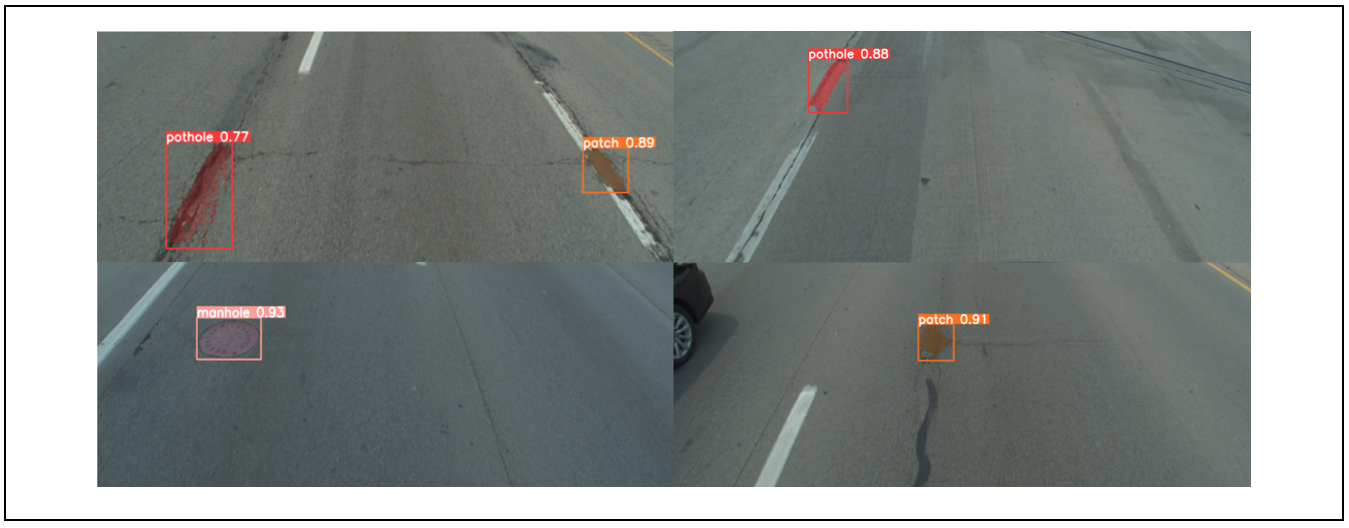


Figure 15. Pothole model detections and segmentations.



Figure 16. Line lane marking model confusion matrix.

confusion matrix, represented in Figure 16, illustrates the model's performance when tested using 143 images. Yellow lane marking was accurately detected by the model in 87% of the cases, with an undetected rate of 10%, and a misidentification rate of approximately 3% as white lane marking. As for white dashed markings, they were successfully identified 70% of the time, misclassified as white lane marking 15% of the time because of the high resemblance between the two, and remained undetected 15% of the time. With regards to white continuous lane marking, the model detected it correctly 73% of the time, misclassified it as white dashed marking 10% of the time and as yellow lane marking 2% of the time. The white continuous lane marking remained undetected 15% of the time. The lack of background images



Figure 17. Line lane marking model detections and segmentations.

during the model's training led to a high percentage of false positives. Additionally, the limited number of training instances resulted in false negatives across the three classes during testing. Figure 17 depicts the model detection of the three classes.

Camera Calibration

Two area maps with sizes similar to the captured images were generated following the calibration of the camera using the two approaches. Each coordinate in these maps represents the real area in square inches, corresponding to the pixel of the image at the same coordinate. The board utilized in the second phase of the experiment with an area of 194 square inches was used to assess the precision of the estimated area from both maps. This board was positioned at varying locations within the camera's field of view, and an image was captured at each position. Subsequently, error analysis was conducted using these images, where the actual area of the board was compared with the area estimated by each approach. As illustrated in Figure 18, the distribution of error for each approach demonstrates differing levels of accuracy. The

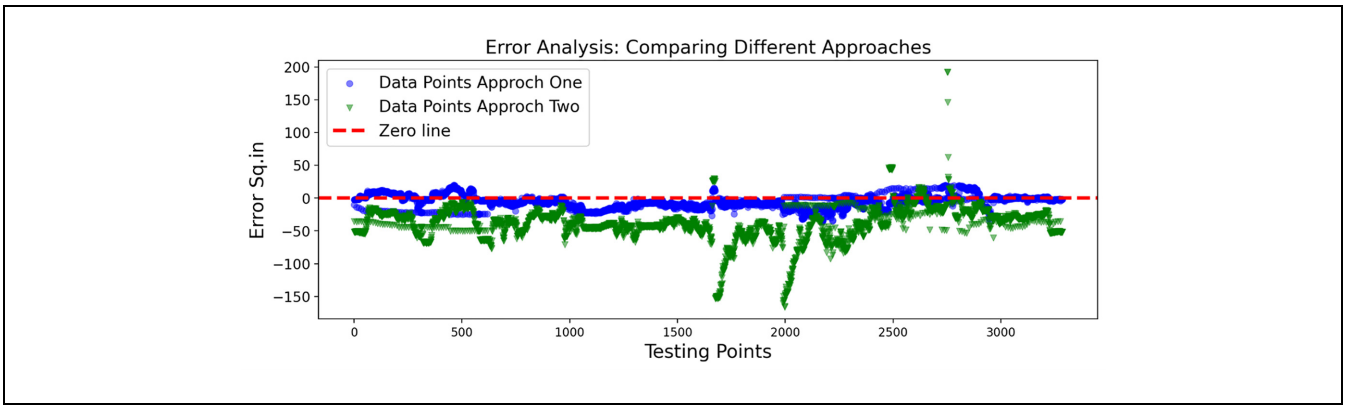


Figure 18. Error analysis comparing the two approaches.

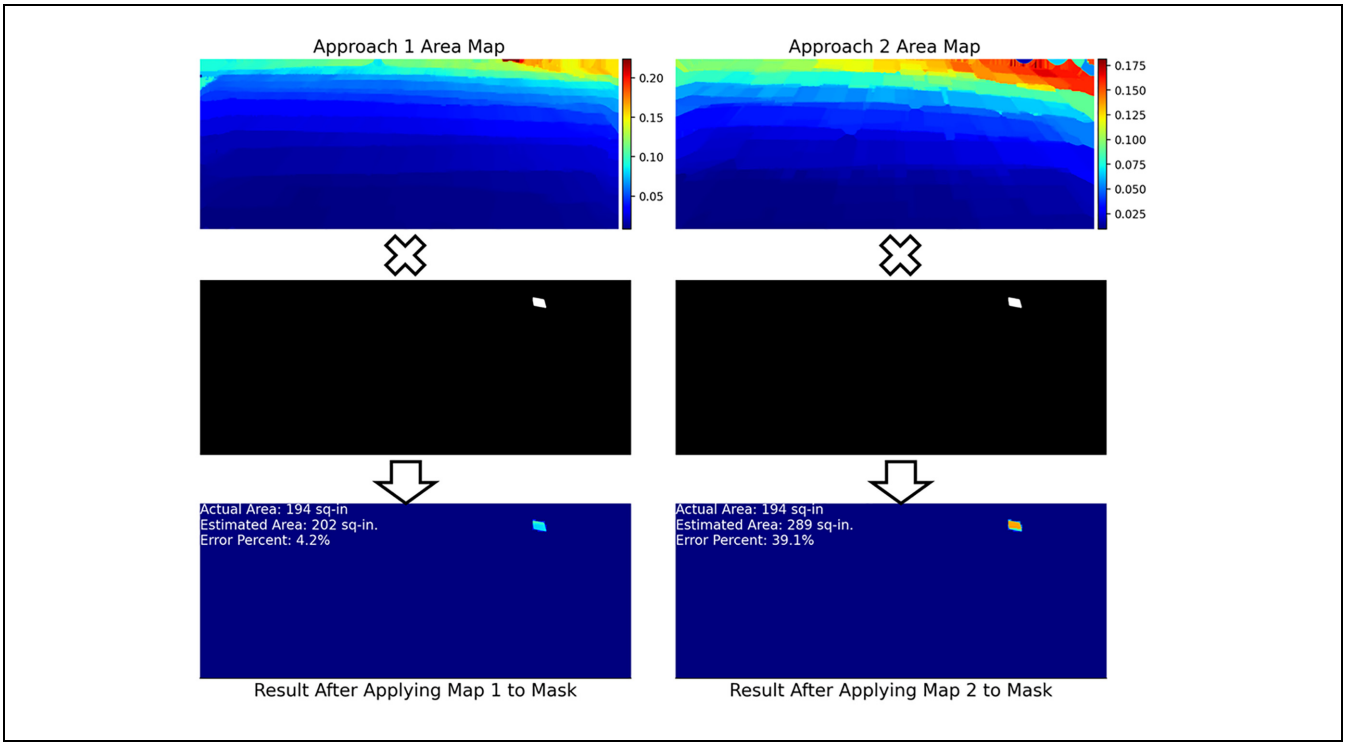


Figure 19. The process of estimating area from the area maps.

higher error associated with the second approach is evident, as most of its estimations are noticeably far from the zero line. Conversely, the first approach exhibits values closer to the zero line, indicating a more accurate estimation.

To obtain the area of a segmented object in an image, the image is first converted to binary format, where "1"s represent the segmented object. Afterwards, the binary image is multiplied element-wise with the area map, and the sum of this multiplication yields the estimated area in square inches. Figure 19 elucidates this process using the two area maps. The color bar in both maps denotes the ranges of values in square inches held by each coordinate of the map. A notable abnormality is apparent in the upper right part of the area map associated with the second approach, where certain coordinates within the same region display both low and high values. In contrast, the area map of the first approach exhibits a higher level of consistency and better distribution of the area. In the final system, the first approach is employed to determine the area of the detected potholes.

Incorporating Models

During the system deployment the outputs of the instance segmentation models designated for both pothole and line lane marking detection are processed separately. Subsequently, the results from both models are

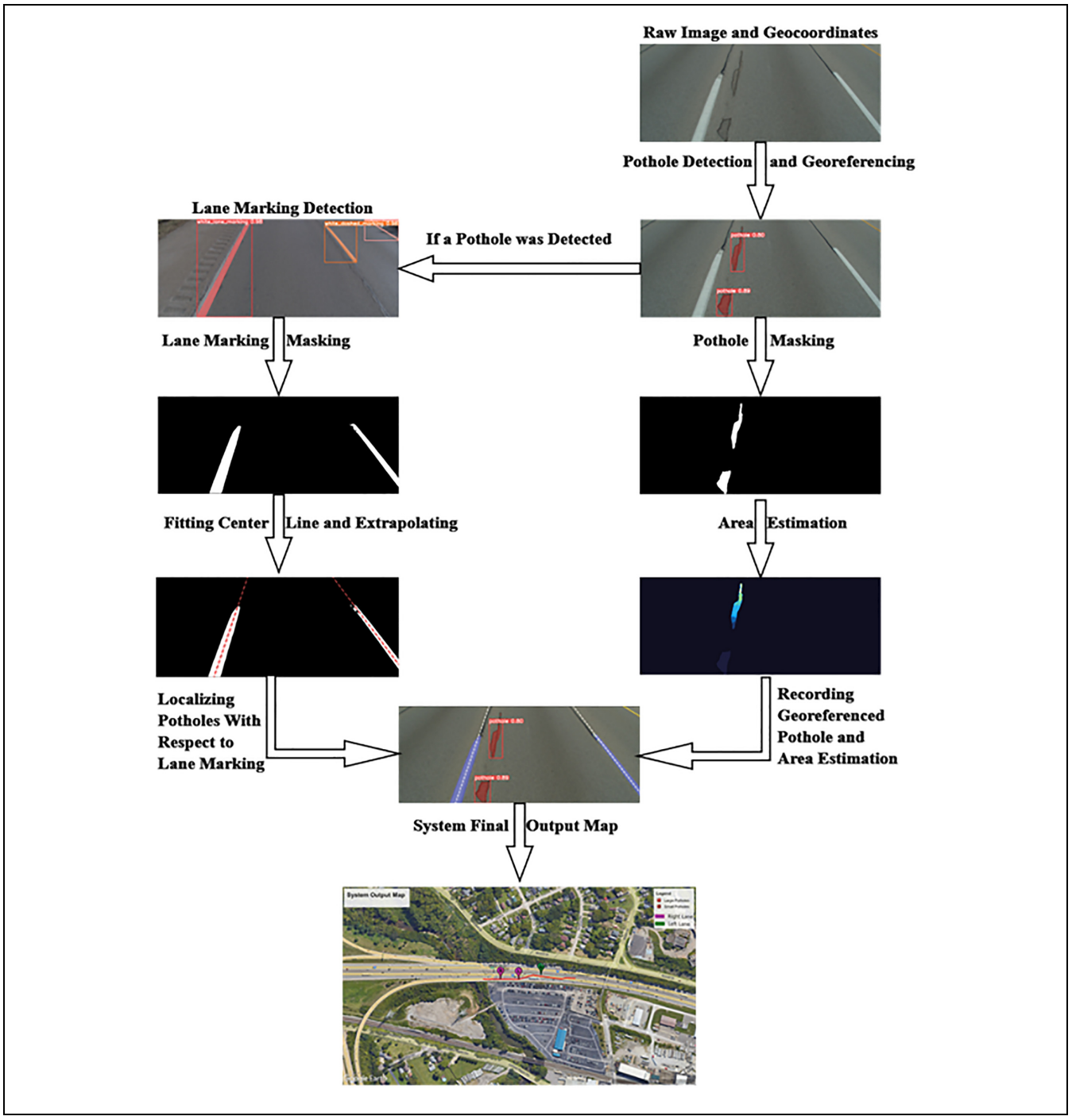


Figure 20. System process to produce the final output map.

integrated to create a map. This map incorporates the geolocation of the detected potholes, the size category of each pothole (small or large), and the location of the pothole with respect to the pavement lanes—whether it is located on the left, right, or center lane. The flowchart in Figure 20 illustrates the system's process for producing the final output. Initially, the raw image obtained by the camera, along with its corresponding geocoordinates from the GPS, are input into the pothole detection model. If a pothole is detected within this image, it is passed to the lane line marking model to detect and segment the marking present in the image. Following this step, each model's output is further processed. For the potholes that have been segmented, the masks are extracted, and the corresponding pothole areas are estimated using the area map. For segmented lane line

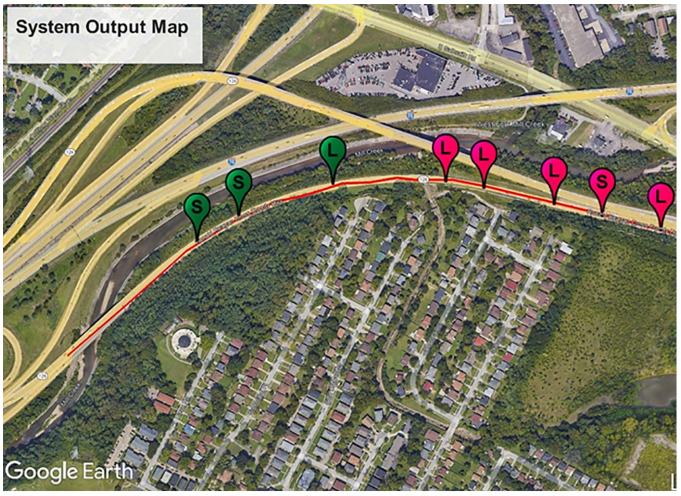


Figure 21. Output map of the system.

markings, the masks are extracted, and a line is fitted through each lane line marking. In the final phase, the pothole's geolocation, estimated area, and location in relation to the pavement lanes are recorded. A detailed map that incorporates all of this information is then generated. Figure 21 depicts the output map of the system.

A notable aspect of this system is its time efficiency in processing images. The algorithm processes each frame in approximately 25 ms, enabling real-time identification of potholes in evaluated roads. This processing speed is crucial, especially when compared with manual identification by a human rater. During the testing phase, human raters missed some potholes that the system accurately identified, underscoring the system's superior efficiency and reliability in real-time object detection in high-speed environments. Thus, the system may potentially empower transportation agencies to prioritize maintenance according to the count and area of potholes. With the precise geolocation, size categorization, and lane location of each pothole, authorities might be able to devise efficient and targeted maintenance plans ahead of time.

Thus, the information that the system provides could assist transportation agencies in determining the most suitable crew size for each section, thus enhancing the productivity and efficacy of maintenance tasks. Beyond labor force optimization, such preemptive planning could offer strategic advantages for traffic control. Being aware of which lanes house potholes and therefore need maintenance, authorities might be able to plan for lane closures ahead of time. This advance planning could enable more effective traffic flow management, guaranteeing road safety, reducing delays, and promoting a more orderly, streamlined transportation environment.

Summary and Conclusions

In this study, an effective low-cost system was developed for real-time detection, localization, and area estimation of potholes. The acquisition of a comprehensive training dataset from diverse locations and times, coupled with the development of intelligent algorithms to facilitate data annotations, played a significant role in improving the performance of the system, achieving a mAP of 93% in distinguishing between potholes, manholes, and patches. Systematic calibration of the camera and meticulous training of the YOLOv8 instance segmentation model resulted in an optimized system capable of rapid, precise detection with minimal error in estimating pothole area. A notable aspect of this system is the real-time identification of potholes in evaluated roads. Furthermore, the inclusion of a line lane marking segmentation model enabled a precise localization of potholes with respect to pavement lanes.

The findings of this study demonstrate that the developed system possesses significant potential for real-world application, particularly in aiding transportation agencies. Its implementation is expected to facilitate a more efficient approach in road maintenance practices. By providing precise information on the location and area of potholes, the system enables transportation agencies to better strategize their pothole patching operations along specific routes. This capability is crucial for optimizing resource allocation, including determining the appropriate crew size, specifying necessary equipment, and estimating the required amount of patching material. Additionally, the system aids in the prioritization of road repairs, resource distribution, advance planning for lane closures, and overall enhancement of maintenance workflows.

Certain limitations were identified, which provide direction for future research. Detecting potholes filled with water remains a challenge for image-based detection systems and needs to be addressed in subsequent studies. Additionally, because the current system configuration effectively covers two lanes to maintain the desired image resolution, it is primarily suitable for roads with three lanes or fewer. Therefore, future studies should aim to optimize the system for a wider field of view, thereby enhancing pothole localization on larger roads, without compromising detection precision. Moreover, while this paper concentrated on the detection of potholes, it is recommended that future studies expand the scope of this approach to assess other types of pavement distress.

Author Contributions

The authors confirm contributing to the paper as follows: study conception and design: M. Nazzal, and D. Manasreh; data collection: Y. Matouq, D. Manasreh; analysis and interpretation of results: M. Nazzal, D. Manasreh, Y. Matouq; draft manuscript preparation: Y. Matouq, D. Manasreh, M. Nazzal. All authors reviewed the results and approved the final version of the manuscript.

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ORCID iDs

Dmitry Manasreh https://orcid.org/0000-0002-3083-6242
Munir D. Nazzal https://orcid.org/0000-0002-5677-1280

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